

Hands-On-Learning: Endpoint Visibility with Sysmon & EDR Pipeline

Learning Objectives

By the end of this activity, learners will be able to:

- Install and configure **Sysmon** to generate endpoint logs.
 - Observe how Sysmon captures **process, file, and network events**.
 - Understand how collected logs fit into the **EDR pipeline (Collect → Detect → Respond)**.
 - Document findings for peer review.
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Structure of the Activity

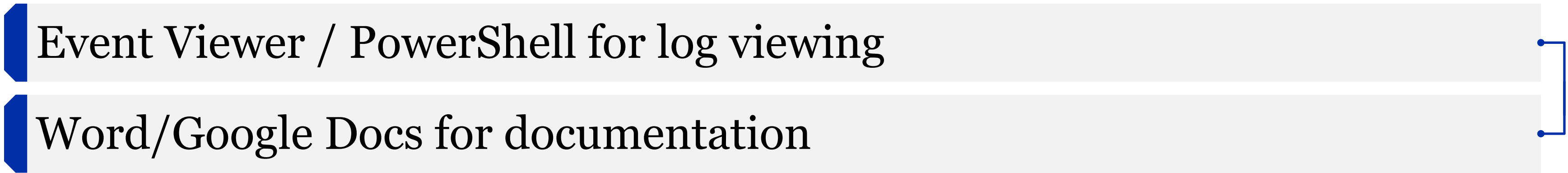
Title: Using Sysmon to Enhance Endpoint Detection

Tool used:

Windows 10/11 VM (recommended for lab safety)

Sysinternals Sysmon (<https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>)

Sysmon config file (example, from [SwiftOnSecurity Sysmon config](#))



Event Viewer / PowerShell for log viewing

Word/Google Docs for documentation

Scenario

Your organization recently faced a malware incident where a suspicious script executed and made outbound network connections. To improve future detection, you are tasked to deploy Sysmon and analyze how endpoint logs reveal malicious activity.

Objective

- Deploy Sysmon.
- Generate and observe logs for **process creation and network connections.**
- Map the collected data to the EDR pipeline stages.



Instructions for Learners

Step 1: Install Sysmon

- 1 Download Sysmon from Microsoft Sysinternals.
- 2 Extract the zip file.
- 3 Open **Command Prompt (Admin)**.
- 4 Install with a baseline config: `sysmon.exe -accepteula -I sysmonconfig-export.xml`.

(If no config file, install with default: `sysmon.exe -accepteula -i`)

Step 2: Confirm Sysmon is Running

Open **Services.msc** → Check if **Sysmon64** service is running.

Open **Event Viewer** → Navigate to: Applications and Services Logs → Microsoft → Windows → Sysmon → Operational.

Step 3: Generate Events (Process Creation & Network)

Open **PowerShell** and run a harmless command: `notepad.exe`.

Check Sysmon logs for Event ID 1 (Process Creation).

Next, run a network command: `curl google.com`.

Check Sysmon logs for Event ID 3 (Network Connection).

Step 4: Analyze the Logs

Identify:

Event ID 1 (Process Creation): Shows executable path, hash, parent process.

Event ID 3 (Network Connection): Shows source/destination IP and ports.

Document findings in a table like:

Event ID	Activity	Example Observed	CIA/EDR Mapping
1	Process Creation	notepad.exe launched	Collect → Detect
3	Network Connection	curl google.com	Collect → Detect

Step 5: Document Your Work

Take **screenshots** of:

Sysmon installation command.

Sysmon logs in Event Viewer.

Explain how each event type fits into the **EDR pipeline (Collect → Detect → Respond)**.

Organize Information Following This Structure for Peer Review

Title Page:

Project Title: “Sysmon for Endpoint Visibility”

Your Name

Date

Requirement 1: Install and configure Sysmon.

Requirement 2: Generate and analyze Sysmon logs.

Requirement 3: Map findings to the EDR pipeline.

Instructions for Documenting and Sharing the Project for Peer Review

Document the Project



Use Word/Google Docs.

Insert screenshots of Sysmon installation & log entries.

Provide a short table mapping Sysmon events to the EDR pipeline.



Share the Project

Save as PDF.

Upload to the “Peer Review” area.

Provide a short description in your submission post.



Peer Review

Review peers' projects.

Check if Sysmon logs were captured correctly.

Comment on whether their mapping to the EDR pipeline makes sense.